Task A1 — Constraint Analysis

LogiEdge: Edge AI cold-chain monitoring | FreightBridge Logistics Pvt. Ltd. | Component A, Module 1

At a glance

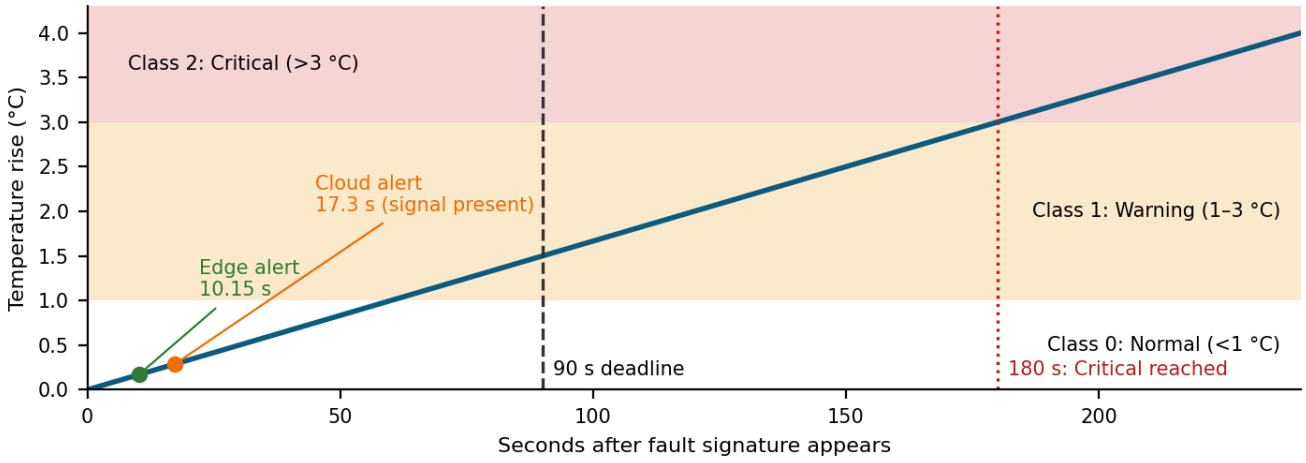
Constraint	Cloud-only system	LogiEdge (edge)	Verdict
Latency (limit 90 s)	17.3 s, only with signal	10.15 s, always	Edge
Data per truck per day	259.89 MB (₹25.99)	0.034 MB (₹0.0034)	Edge, 7,660× less
Connectivity gaps	Blind for 35–90 min	Works offline, syncs later	Edge
Privacy	Raw data leaves truck	Raw data stays on truck	Edge

1 Latency

Given: *a refrigeration fault raises cargo temperature by 1 °C per minute; alert needed within 90 s. Round-trip times assumed 0.2–1 s on rural 4G.*

Step	Working	Result
Rise at deadline	90 s × (1 °C ÷ 60 s)	1.5 °C (Warning band)
Time to Critical (>3 °C)	3 °C ÷ 1 °C/min	180 s
Cloud path	10 + 4.8 + 1 + 0.5 + 1 s	17.3 s (19% of budget)
Edge path	10 + 0.05 + 0.1 s	10.15 s (11% of budget)

Finding: cloud inference is feasible on latency alone, but only while a signal exists. Latency does not rule out the cloud; connectivity does.

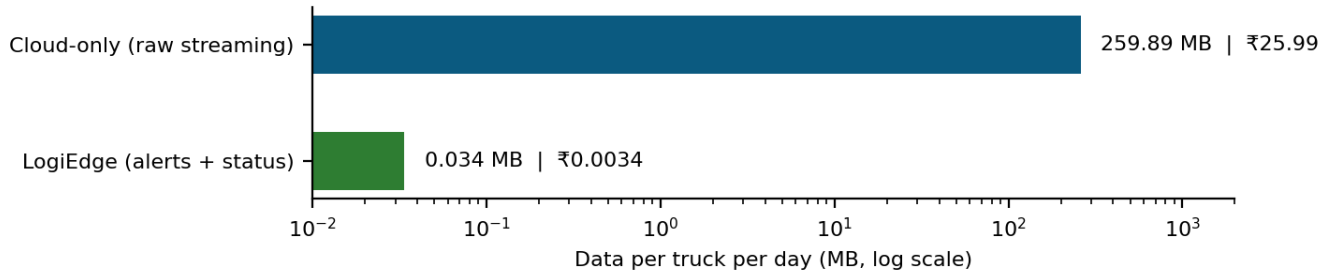


2 Bandwidth and cost

Given: *temperature 1 Hz; vibration 500 Hz × 3 axes; door events discrete; tariff ₹0.10/MB.*

Item	Working	Result
Vibration	500 × 3 × 2 B × 86,400 s	259.2 MB/day
Temperature	8 B × 86,400 s	0.69 MB/day
Door events	50 × 16 B	0.0008 MB/day
Raw total per truck	Sum of the above	259.89 MB/day
Cost per truck	259.89 × ₹0.10	₹25.99/day
Fleet of 85 trucks	259.89 × 85	22.09 GB/day; ₹2,209/day; ≈ ₹8.06 lakh/year
Edge alerts + status	288 × 100 B + 20 × 256 B	0.034 MB/day; ₹0.0034/truck/day; ≈ ₹105/year fleet

Finding: edge processing cuts volume and cost by a factor of 7,660 (99.99%).



3 Connectivity

Given: *seven gaps of 35–90 minutes each, so 245–630 minutes (4.1–10.5 hours) without signal. One gap is 23–60 times the 90 s budget.*

During a gap	Cloud-only	LogiEdge
Inference	None	Continues on the truck
Alerts	None; fault goes unnoticed	In-cab alarm within about 10 s
Data	Not recorded	Encrypted, hash-chained queue plus 1 Hz temperature log; 90 min of raw data is only 16.2 MB
Signal returns	Nothing to recover	Ordered upload with retries and acknowledgements

Finding: a fault at gap entry passes 3 °C within three minutes with nobody informed, which is the Nashik–Aurangabad vaccine-spoilage scenario. The edge design leaves no unmonitored period and keeps chain-of-custody records unbroken.

4 Privacy

- **Data stays local:** 259 MB/day of raw data remains in the cab; only class labels and signed alerts (about 34 KB/day) leave.
- **Encryption:** AES-256 at rest with hardware-backed keys; TLS 1.3 with mutual authentication in transit.
- **Tamper evidence:** device-signed, hash-chained logs make alteration detectable.
- **Contract:** a data-minimisation commitment that raw cargo-condition data is processed and retained only on a key-protected device, with disclosure limited to defined alert fields.
- **Caveat:** device hardening and access policies are still required.

Appendix A1 — Assumptions and Workings

Decimal units: 1 MB = 1,000,000 bytes; 1 lakh = 100,000. “Assumed” values are engineering estimates, not FreightBridge measurements.

A. Assumptions

Item	Value	Basis
Monitoring period	86,400 s/day (24 h)	Assumed; refrigeration runs while parked
Temperature record	8 B (4 B float32 + 4 B timestamp)	Assumed
Vibration sample	2 B (16-bit) per axis; timestamp implicit	Assumed
Door events	50 per day × 16 B	Assumed
Edge status packet	100 B every 5 min (288 per day)	Assumed
Edge alert message	256 B signed JSON; 20 per day	Assumed, conservative
Rural 4G round trip	1 s (range 0.2–1 s; 2G fallback up to about 3 s)	Assumed typical
Effective uplink	50 kbps = 6,250 B/s	Assumed
Feature window	10 s	Design choice
Edge inference / local alarm	0.05 s / 0.1 s	Assumed
Cloud inference / alert push	0.5 s / 1 s	Assumed
Tariff	₹0.10 per MB	Given

B. Latency workings

Step	Calculation	Result
Rise per second	1 °C ÷ 60 s	0.0167 °C/s
Uplink payload	10 s × 3,000 B/s	30,000 B
Upload time	30,000 B ÷ 6,250 B/s	4.8 s
Cloud total	10 + 4.8 + 1 + 0.5 + 1	17.3 s; 17.3 ÷ 90 = 19.2%
Edge total	10 + 0.05 + 0.1	10.15 s; 10.15 ÷ 90 = 11.3%

C. Bandwidth workings

Step	Calculation	Result
Vibration rate	500 samples/s × 3 axes × 2 B	3,000 B/s
Vibration per day	3,000 × 86,400	259,200,000 B = 259.2 MB
Temperature per day	8 × 86,400	691,200 B = 0.6912 MB
Raw total per truck	259.2 + 0.6912 + 0.0008	259.892 MB
Cost per truck	259.892 × 0.10; × 365	₹25.99/day; ₹9,486/year
Fleet volume	259.892 × 85	22,090.8 MB = 22.09 GB/day
Fleet cost	22,090.8 × 0.10; × 365	₹2,209.08/day; ₹806,315/year
Edge status	288 × 100 B	28,800 B
Edge alerts	20 × 256 B	5,120 B
Edge total	28,800 + 5,120	33,920 B = 0.03392 MB/day
Edge cost per truck	0.03392 × 0.10	₹0.0034/day

Step	Calculation	Result
Edge fleet cost	0.03392 × 85 × 0.10; × 365	₹0.288/day; ₹105.24/year
Reduction	259.892 ÷ 0.03392; 1 − (0.03392 ÷ 259.892)	7,662 times; 99.987%

D. Connectivity workings

Step	Calculation	Result
Total dark time	7 × 35; 7 × 90	245 min (4.08 h); 630 min (10.5 h)
Gap vs. budget	2,100 s ÷ 90; 5,400 s ÷ 90	23.3 times; 60 times
Raw buffer, 90 min	3,000 B/s × 5,400 s	16,200,000 B = 16.2 MB

E. Items to confirm

- Bytes per sensor sample and timestamp format; vibration sensor resolution.
- Daily door-event count and expected alert count.
- Measured round-trip times and uplink speeds on the Nashik–Aurangabad route.
- Whether 35–90 minutes applies to each of the seven locations (as calculated) or to the route in total, since 4.1–10.5 hours is long for that journey.
- Whether monitoring runs 24 hours a day.